Firmware updates over LoRa — a combined approach

Proposal for discussion, 7 November 2026

Summary

Three directions are on the table: a tested page transfer through a relay, delta updates with a split firmware layout, and campaigns run either automatically or step by step by an operator. They do not compete. They are layers of one system, built on the same target behaviour: check a signed package, accept a signed profile switch, receive, verify, activate, confirm.

One decision cannot wait: the partition table. It cannot be changed over the air, so the layout installed at the next site visits is the layout those sites keep.

What is tested today

Bench, TTGO LoRa32 V2.1 with SX1278, controller → one relay → target, same site.

Function	Status
Transfer in 4 KB pages, bitmap acknowledgements, repair cache in the relay	Validated: complete image through one relay
Resume from flash after a target reset, a relay restart or a power cut	Validated on short transfers
Temporary fast profile (BW500/SF7) by signed command, automatic return when its lease expires	Validated
Activation as a separate signed command, radio confirmation after boot, rollback otherwise	Validated (uncompressed images)
Relay and target routes set by signed APRS commands (`0TA=RELAY,<target>`, `0TA=VIA,<relay>`)	Validated: both routes set by radio, routed transfer started with the new route key, 241 KB written then interrupted on purpose
Image installed locally (web page or USB) accepted at boot; radio-activated images still wait for the radio confirmation	Validated on the relay board
Compressed pages: each 4 KB page deflated on its own, decoded on the target before writing	64 KB sample through the relay, decoded SHA-256 verified on the target; full image 27% smaller (1,298,854 instead of 1,778,688 bytes)

Not tested: activation of a compressed image, ESP32-S3 / SX1262 builds (they compile), more than one relay, range, a busy APRS channel.

Layers of the combined approach

Layer	Proposal	Status
Transport to one station	The page transfer above	Tested on the bench

Layer	Proposal	Status
Payload	Delta (detools/bsdiff, then xz with an 8 KB dictionary) for the application; codec and model sent whole; compressed pages as the fallback for a complete image	Delta sizes measured on real builds; not run on a board
Firmware layout	Application, model and codec in separate partitions, each with an A and a B copy	To be agreed
Authorisation	Packages signed by the operator; profile switches requested only by authorised distributors	To be designed
Several stations at once	Broadcast with forward error correction, beside the acknowledged transfer	Later

Measured patch sizes

Change between two builds	Patch	Tool
Web page label only	46.5 KB (2.6% of the image)	xdelta3
Page decompression added	203 KB (11.3%)	xdelta3
17 lines added	20.6 KB	detools + xz
Whole-text PPM to LACF 2.0	302 KB	detools + xz

Most of a patch comes from code that moves: the same source built twice differs by 495 bytes. Keeping the model and the codec outside the application is what keeps format changes small.

Two ways to run a campaign

Both use the same firmware. Only the sender of the commands changes.

Piloted. The operator updates a first digipeater, then tells it to serve a station it hears well and the operator does not. During the transfer the serving digipeater returns briefly to the normal profile to report progress, repairs and errors. The operator decides each next step. When both stations use the same board, the serving digipeater can send its own confirmed image from its running partition, with no extra storage.

Automatic. An iGate fed by MQTT or Wi-Fi receives a signed package and a campaign: stations, time windows, and SNR thresholds taken from RXT tuples and the neighbour table (SF7, SF9 or SF10 depending on the link). It proceeds on its own and reports.

Safeguards for automatic campaigns:

- each station opts in by a setting;
- a limit on the number of digipeaters off the APRS channel at the same time;
- stop at the first error;
- operator pause and abort at any time;
- the boot confirmation and rollback stay as they are.

Two separate rights

Providing a package (signed by the operator) and requesting a profile switch (authorised distributors) are separate. A digipeater or an iGate can then serve an image without holding the target’s secret,

and without being able to produce one.

The partition table

Proposed application slots are 1,664 KB each. The current update-capable build is 1,785,765 bytes, so it fits only once the model and the codec are moved out of the application; the application alone is estimated, not measured, at about 1,450 KB. The sizes, the model limit and the module interface have to be agreed before the first release that uses them, and before sites that will be out of reach receive firmware.

Proposed order

1. Agree the partition layout and the model limit.
2. Signed package manifest; HELL0 reports the SHA-256 of the running application, so a delta can check its base.
3. Delta applied in order into the inactive slot, on the page transport, with resume by patch position.
4. Piloted store-and-forward, with status reports on the normal profile.
5. Broadcast with forward error correction, and automatic campaigns.

Open points

- Partition sizes, model limit and the codec module interface.
- Signature scheme and key handling.
- Status report interval during a transfer, and its cost in transfer time.
- SNR thresholds for choosing SF7, SF9 or SF10.
- Field trials: accessible sites first, never a site that will be out of reach.